

Searching 2: Binary Search Problems

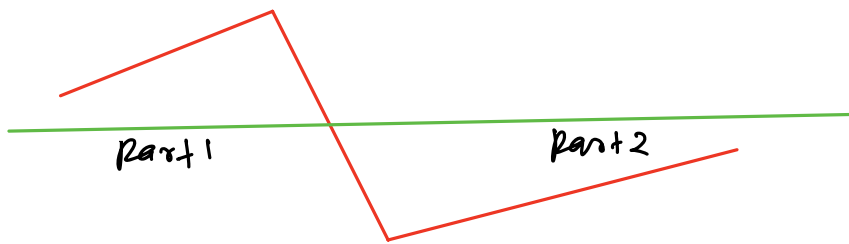
3 steps of BS

1. Define search space
2. find mid element & check if it the answer
3. Decide whether to go left or right

Question 1

Given a sorted rotated array of unique elements.
find index of given element K.

$A = [\overset{0}{6} \overset{1}{7} \overset{2}{8} \overset{3}{1} \overset{4}{2} \overset{5}{3} \overset{6}{4} \overset{7}{5}]$



find the smallest element?

How to if $A[mid]$ is in Part 1 or Part 2?

$A[0] >$ all elements in Part 2

$A[0] <$ all elements in Part 1

check for not rotated? $\rightarrow A[0] < A[n-1]$

↓
 $TC = O(\log N)$ $SC = O(1)$

Solve in 1 binary search

// Define search space

$l = 0, \quad r = n - 1$

while ($l \leq r$) {

 // check if mid is answer !

$mid = l + (r - l) / 2$

 if ($A[mid] == K$) return mid

 // decide whether to go left or right

 if ($K < A[0]$) { // K is in Part 2

 if ($A[mid] < A[0]$) { // mid is in Part 2

 if ($A[mid] < K$) $l = mid + 1$

 else $r = mid - 1$

 }

 else { // mid is in Part 1

$l = mid + 1$

 }

}

else { // K is in Part 1

K	mid	
Part 1	Part 2	go left
Part 1	Part 1	regular BS
Part 2	Part 2	regular BS
Part 2	Part 1	go right

if ($A[mid] > A[0]$) { // mid is in Part 1

if ($A[mid] < K$) $l = mid + 1$

else $r = mid - 1$

}

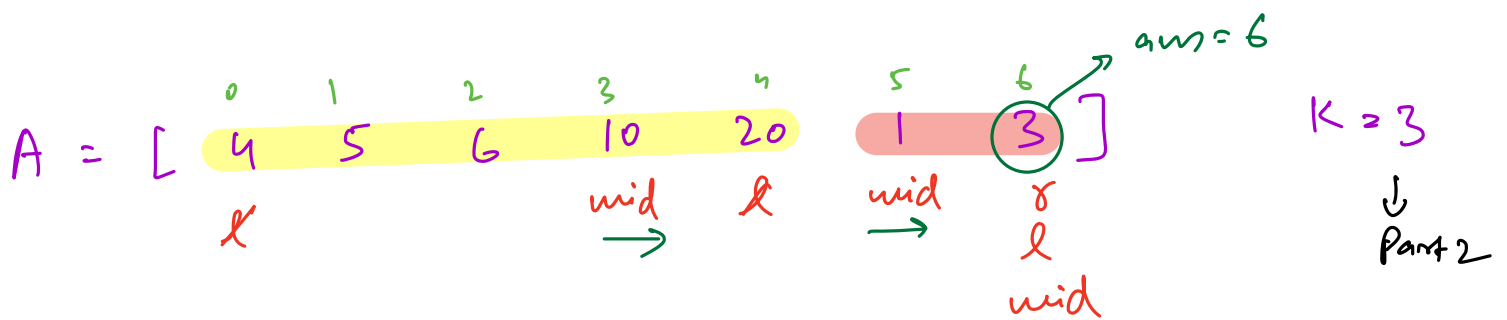
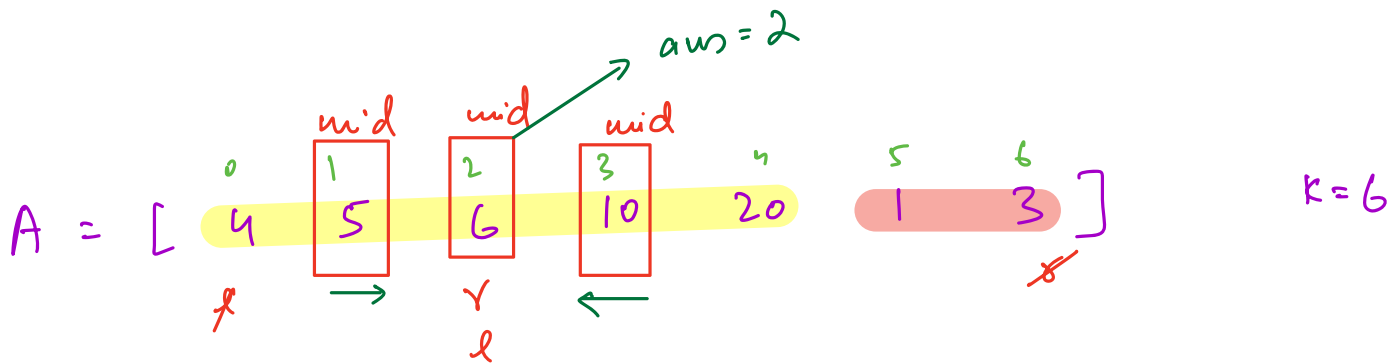
else { // mid is in Part 2

$r = mid - 1$

}

}

return -1



TC = $O(\log N)$

SC = $O(1)$

HW $\rightarrow$ try in rotated descending array

Question 2

find sqrt. of a given perfect sq N . (w/o inbuilt functions)

$$N = 49 \quad \text{ans} = 7$$

$$N = 25 \quad \text{ans} = 5$$

$$~~N = 30~~ \quad \longleftarrow \text{invalid input}$$

Brute force

min value = 1
max value = N } range

```
for ( i = 1 to N ) {  
    if ( i*i == N )  
        return i  
}
```

}

$$TC = ~~O(N)~~ \quad SC = O(1)$$

$$O(\sqrt{N})$$

$\sqrt{N}$ iterations

$$N = 25 \quad i = 1, 2, 3, 4, 5$$

$$49 \quad i = 1, 2, 3, 4, 5, 6, 7$$

$$81 \quad i = 1, 2, 3, \dots, 9$$

// Define search space

$$l = 1, \quad r = N$$

```
while ( l <= r ) {
```

// check if mid is answer!

$$\text{mid} = l + (r - l) / 2$$

if (mid * mid == N) return mid

// decide whether to go left or right

if (mid * mid < N) l = mid + 1 // go right

else r = mid - 1

}

TC = $O(\log N)$ SC = $O(1)$

$N = 36$

$l = 1$ 5

$r = 36$ 17 8

$mid = 18$ 9 4 6

$18^2 > 36$

$9^2 > 36$

$4^2 < 36$

$6^2 = 36$
↑
ans

mid * mid == N
may overflow

$\Rightarrow mid == N / mid$

$$1. \gcd(a, b)$$

$$TC = O(\log(\max(a, b)))$$

$$2. \text{lcm}(a, b)$$

$$\text{lcm} = \frac{a \times b}{\gcd(a, b)}$$

$$\text{lcm}(10, 15) = 30$$

$$\text{lcm}(6, 10) = 30$$

Question 3

find the N^{th} number which is divisible by either X or Y or both.

$$N=5$$

$$X=2$$

$$Y=3$$

$\rightarrow 2, 3, 4, 6, \overset{\text{5th no.}}{\textcircled{8}} \text{ ans}$

$$N=7$$

$$X=6$$

$$Y=10$$

$\rightarrow 6 \quad 12 \quad 18 \quad 24 \quad 30 \dots\dots\dots$

$\rightarrow 10 \quad 20 \quad 30 \quad 40 \quad 50 \dots\dots\dots$

6 10 12 18 20 24 $\textcircled{30} \text{ ans}$

Brute force

cut = 0 i = 0

while (cut < N) {

 i++

 if (i % X == 0 || i % Y == 0)

 cut++

}

return i

Only iterate over the multiples of X & Y

N = 8

X = 6 → ~~6~~ ~~12~~ ~~18~~ ~~24~~ ~~30~~ ~~36~~ 42

Y = 10 → ~~10~~ ~~20~~ ~~30~~ ~~40~~ 50 60 70

merge 2 sorted into
1 sorted array

6 10 12 18 20 24 30 (36)

TC = O(N) SC = O(N) → HW in SC = O(1)

Optimization

of multiples of $X \leq K \rightarrow K/X$

$$\begin{array}{l} X=6 \\ K=50 \end{array} \rightarrow \frac{50}{6} = 8$$

6, 12, 18, 24, 30, 36, 42, 48
8 multiples

of multiples of X or Y or both $\leq K$

$$\frac{K}{X} + \frac{K}{Y} - \frac{K}{\text{lcm}(X,Y)}$$

$$\begin{array}{l} X=6 \rightarrow 50/6 = 8 \\ Y=10 \rightarrow 50/10 = 5 \end{array} \quad \left. \begin{array}{l} 8 \\ 5 \end{array} \right\} 13 - 1 = 12$$

$$K=50$$

6 12 18 24 30 36 42 48

10 20 30 40 50

$$lcm = x * y / gcd(x, y)$$

// define search space

$$l = \min(x, y) \quad r = N * \min(x, y)$$

while (l <= r) {

// check if mid is answer

$$mid = l + (r - l) / 2$$

$$cut = \frac{mid}{x} + \frac{mid}{y} - \frac{mid}{lcm}$$

multiples of x or y
or both <= mid

if (cut == N && (mid % x == 0 || mid % y == 0))

return mid

// decide whether to go left or right

if (cut < N) l = mid + 1

else r = mid - 1 // cut >= N

}

$$N = 8$$

$$x = 6$$

$$y = 10$$

$$mid = 37 \text{ x}$$

$$\frac{37}{6} = 6 \quad \frac{37}{10} = 3$$

$$\frac{37}{30} = 1$$

$$6 + 3 - 1 = 8$$

$$X = 6 \quad \text{sum} = 30$$

$$Y = 10$$

$$N = 7$$

l	r	mid	cut
6	42	24	$\frac{24}{6} + \frac{24}{10} - \frac{24}{30} = 4 + 2 - 0 = 6$
25	42	33	$\frac{33}{6} + \frac{33}{10} - \frac{33}{30} = 5 + 3 - 1 = 7$
25	32	28	$\frac{28}{6} + \frac{28}{10} - \frac{28}{30} = 4 + 2 - 0 = 6$
29	32	30	$\frac{30}{6} + \frac{30}{10} - \frac{30}{30} = 5 + 3 - 1 = 7$

$$\text{am} = 30$$

$$TC = O(\log(N + \min(X, Y))) \quad SC = O(1)$$



$$\frac{r-l+1}{2 \times 2 \times 2} \Rightarrow 1$$

$$(r-l+1) = 2^x$$

$$x = \log(r-l+1)$$

total iterations

Median in an array

middle element when array is sorted.

if odd no. of elements = $a[n/2]$

even no. of elements = $\frac{a[n/2 - 1] + a[n/2]}{2}$

Question 4

find median of two sorted arrays

$A = [1 \ 3 \ 5]$

$B = [2 \ 4 \ 6]$

$[1 \ 2 \ 3 \ 4 \ 5 \ 6]$

$$\frac{3+4}{2} = 3.5$$

$A = [1 \ 2 \ 3]$

$B = [10 \ 20 \ 30]$

$[1 \ 2 \ 3 \ 10 \ 20 \ 30]$

Idea 1: Merge & find median

$$TC = O(N+M)$$

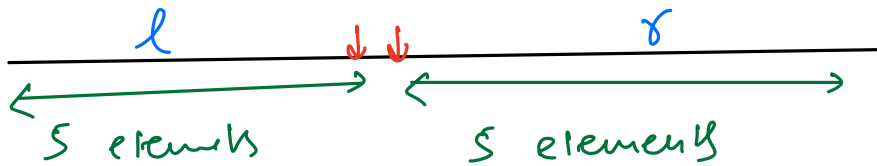
$$SC = O(N+M)$$

Idea 2 : Partitioning

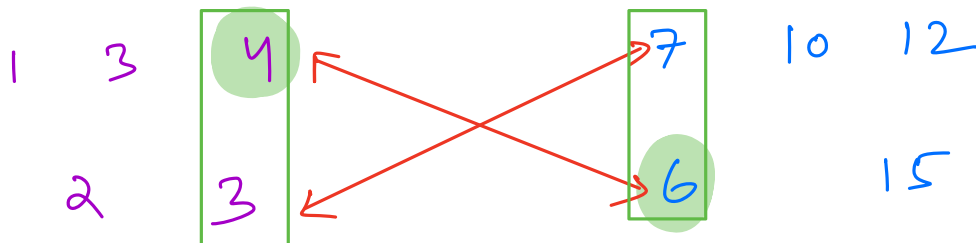
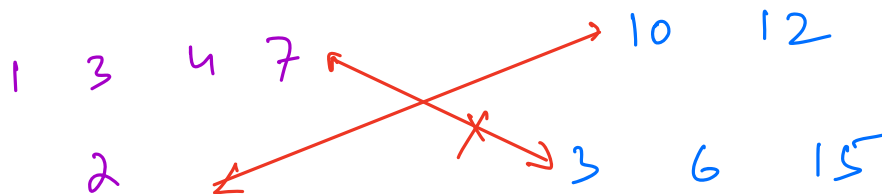
$$A = [1 \ 3 \ 4 \ 7 \ 10 \ 12] \quad N=6$$

= 10 elements

$$B = [2 \ 3 \ 6 \ 15] \quad M=4$$



$$l \leq r$$



$$\frac{4+6}{2} = 5$$

To be continued